DataTürk

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***Abstract*—This document is the first progress report for DataTürk's machine learning project. The document contains the details of the dataset cleansing process and the summary of the given dataset.**

***Index Terms*—machine learning, github, dataset, missing values, graphs**

I. Introduction

We have created a python script [1], in order to identify the missing values in our given dataset, and analyze the data according to their contextual meaning. Later, we visualised some parts of the dataset to be able to understand it better.

II. Data Preprocessing

A. Dropping Some of the Columns and Filling the Values

We identified the columns that contained more than 50 percent of the missing data values in them and filled in these numeric columns using the calculated mean of such columns [2]. For non-numeric columns, we used the mode. In order to eliminate the bugs in the code, we took advantage of some sources [3] [4] [5].

B. Creating a Binary Label for the Dataset

We converted the last columnt to a binary label column, which gave us a way to tag and identify these columns.

III. Appendices

A. Summary of the Dataset

The dataset is can be seen as:

Statistic	**Value**
Missing values (before)	7318445
Missing values (after)	0
Non-numerical columns	13
Numerical columns	68
Total Rows	246008
Total Columns	81

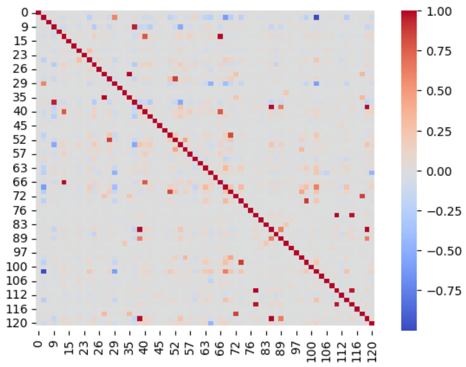

Fig. 1. Correlation between information on the dataset.

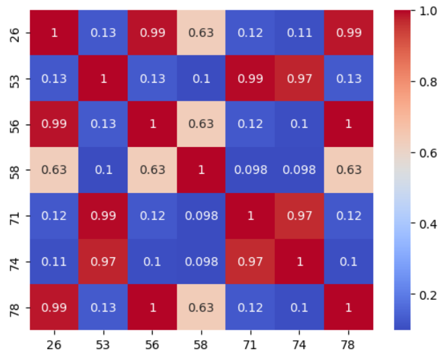

Fig. 2. Heatmap.

Correlation analysis was performed in the first figure we came across. Our main purpose while doing this analysis is to determine the relationship between the variables and to understand them according to their directions and intensities. In this statistical method, it can be said that there is a significant relationship between both variables if it is less

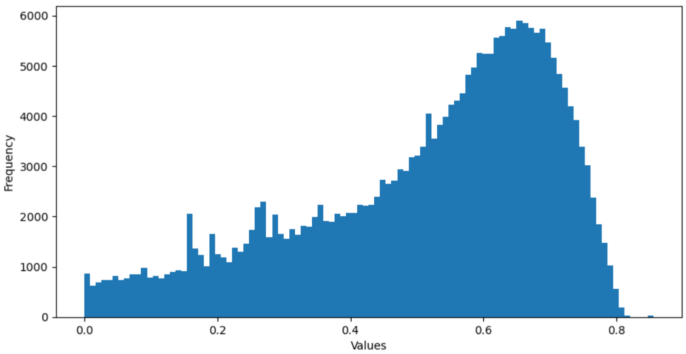

Fig. 3. Histogram of Column 15.

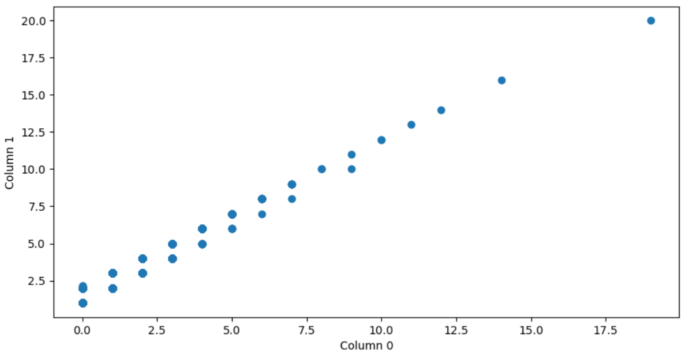

Fig. 4. Positive correlation between the column 50 and 64.

than 0.05 among the meanings we can understand. In our analysis, we proved this again in the heat map above, where the number values are the same in the red visible line. The reason for using heatmap is actually to see the losses, separations or densities more clearly again.

IV. Conclusion and Future Work

Our next step will be to go on the depths of machine learning with our cleansed dataset.

References

[1] iamemirsahin. DataTürk. [Online]. Available: https://github.com/iamsahinemir/DataTurk

[2] Cook, A. Handling Missing Values. [Online]. Available: https://www.kaggle.com/code/alexisbcook/handling-missing-values

[3] Güneş, A. Python ile Veri Temizleme ve Ön İşleme. [Online]. Available: https://alprgunes.medium.com/

[4] TirendazAkademi. Nasıl Veri Bilimcisi Olunur. [Online]. Available: https://tirendazakademi.medium.com/

[5] OpenAI. ChatGPT-3. [Online]. Available: https://chat.openai.com/